Signal-Aware Framework for Multilingual Language Model Evaluation*

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Abstract

Training multilingual language models requires repeated decisions about data mixtures and hyperparameters, typically guided by expensive and redundant benchmark suites that exhibit high variance, particularly in multilingual settings. We propose a framework to identify which benchmarks, and which benchmark subsets, reliably preserve multilingual model rankings across model sizes and training checkpoints. We pretrain 36 multilingual models with varied data mixtures and sizes ranging from 175M to 1B parameters, and extend our analysis up to 70B parameters using open-source multilingual models. We provide a generalizable language-based benchmark reliability map for pretraining evaluation, covering an initial set of 12 typologically diverse languages, enabling cheaper, better-informed multilingual model development.

Keywords

LLMs, Multilingual, Evaluation

1. Introduction

Pretraining multilingual language models requires billions of tokens and hundreds of thousands of euros in compute. Modern post-training projects iterate over dozens of recipes [1], each evaluated on long, expensive benchmark suites. Benchmarks inform decisions that impact the final model quality, but they vary in their diagnostic value. Some are sensitive to recipe changes, others have high variance, redundancy, or weak correlation with downstream objectives. Improvements on certain benchmarks, therefore, may not reflect meaningful progress. Evaluation cost scales with the number of decisions made, and unreliable benchmarks waste a large share of the development budget. In particular, long-context extension produces effective windows shorter than the advertised ones [2], and the later post-training stages supervised fine-tuning (SFT), direct preference optimization (DPO), and Reinforcement Learning with Verifiable Rewards (RLVR) often score open-ended tasks with biased LLM judges [3, 4]. Moreover, these reliability gaps widen for multilingual settings [5, 6, 7].

Recent work has begun to examine the diagnostic value and efficiency of language model benchmarks, recognizing that not all benchmarks provide equally informative signals for guiding model development. Heineman et al. [8] introduce a signal-to-noise ratio (SNR) framework that quantifies benchmark reliability for English models. Their empirical analysis shows that benchmarks with higher SNR more consistently preserve model rankings across scales, improving the accuracy of training decisions and reducing error when extrapolating performance using scaling laws. This work highlights that benchmark choice directly affects the reliability of decisions made during model development. Other work targets evaluation efficiency, identifying benchmark subsets with high discriminative power [9, 10, 11].

Together, these works suggest that benchmark effectiveness depends not only on coverage but also on how much reliable signal a benchmark provides relative to its evaluation cost. However, existing studies primarily focus on English benchmarks and pretraining. Existing decision accuracy metrics

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might fail for the multilingual case: different languages might have different decision accuracy metrics or respond differently to scaling laws. Moreover, the language transfer between languages might affect benchmark selection.

Our goal is to extend the aforementioned frameworks to multilingual settings and identify the most reliable multilingual benchmarks for each training stage that correlate with final model performance. This will enable practitioners to make better, cheaper decisions during multilingual model development.

In particular, the research questions we aim to address during this thesis are:

- Which multilingual benchmarks produce reliable model rankings at small scale, such that those rankings can inform model design decisions during early-stage ablations and generalize to larger models?
- Can benchmark metrics, computed at small scale, serve as a proxy for decision accuracy, removing the need to evaluate large models?
- Which benchmark design choices, including data source and curation, lead to higher decision accuracy?

2. Related Work

Our work draws on three lines of research, namely benchmark reliability for pretraining, multilingual evaluation, and post-training evaluation for RLHF and RLVR.

2.1. Benchmark Reliability

Recent work shows that benchmarks differ substantially in how reliably they guide language model development. Heineman et al. [12] introduce a signal-to-noise ratio (SNR) framework where signal measures a benchmark's ability to separate models and noise captures score variability across checkpoints or stochastic runs. They show that high-SNR benchmarks better preserve rankings and reduce scaling-law prediction error. Complementary work studies benchmark quality through ranking consistency, discriminability, and capability-alignment deviations [13], while IRT-based methods show that many benchmark items have weak discriminative power and can often be replaced by smaller, more informative subsets [14, 15].

Efficiency-oriented work reaches similar conclusions. Gupta et al. [16] propose SMART filtering to remove easy, contaminated, or redundant examples, reducing evaluation size while preserving agreement with independent model rankings. Scaling-law studies further suggest that only some evaluations provide predictable development signal: Chen et al. [17] forecast downstream performance from smaller models, while Schaeffer et al. [18] study scaling laws for generative metrics such as pass@k. Together, these papers motivate our focus on identifying which multilingual benchmarks remain locally reliable across nearby post-training checkpoints, rather than treating all benchmark averages as equally meaningful.

2.2. Multilingual Evaluation

Multilingual benchmarks inherit reliability problems from their English counterparts and add new ones. Most popular benchmarks are machine-translated from English, which introduces translation artifacts, cultural mismatch, and label noise that distort cross-lingual rankings [19, 20, 21]. Tokenizer performance varies by an order of magnitude across languages, inflating both training and evaluation cost for non-Latin scripts and morphologically rich languages [22, 23]. LLM-based judges also lose calibration on lower-resource languages, which further reduces the reliability of post-training evaluation in multilingual settings [6, 7]. These factors make English benchmark reliability metrics insufficient as proxies, and motivate a multilingual-specific framework.

2.3. Reinforcement Learning from Human Feedback

Reinforcement Learning from Human Feedback (RLHF) is a standard post-training paradigm for aligning language models with human preferences. Typical pipelines combine supervised fine-tuning, preference-data collection, reward modeling, and policy optimization under a constraint to a reference model [24, 25]. Recent variants simplify this pipeline. Direct Preference Optimization (DPO) optimizes directly on preference pairs, avoiding explicit reward-model training and online RL [26].

The results of the RLHF pipeline are judged on subjective qualities such as helpfulness, harmlessness, and instruction following. These are qualities that likelihood cannot capture, so evaluation relies on LLM judges and reward models, and both inherit reliability problems. Judges are biased by length and style [3, 4] and lose competence in lower-resource languages [6, 7], and reward models can produce gains that reflect artifacts rather than capability improvements [27]. Benchmark reliability is therefore central for selecting among RLHF recipes, and even more so in multilingual settings.

2.4. Reinforcement Learning with Verifiable Rewards

Reinforcement Learning with Verifiable Rewards (RLVR) instead uses objective validators such as exact-answer checks, unit tests, symbolic solvers, or task-specific verifiers. It is therefore well suited to math, code, and structured reasoning tasks [28, 29, 30, 31].

This shifts the reliability concern from human or judge bias to verifiers. Validators may be sparse, brittle, or exploitable, and final-answer accuracy can confuse robust reasoning with lucky sampling or benchmark-specific optimization [32, 33]. For RLVR, benchmark reliability should therefore measure whether verified scores provide stable, discriminative, and generalizable signal across nearby post-trained models.

3. Methodology

We extend the Signal and Noise framework of Heineman et al. [8] from English to multilingual language model development. We first define decision accuracy (DA) as whether benchmark rankings transfer from smaller to larger models. We propose the signal-to-noise ratio (SNR) as a proxy for decision accuracy, and present multiple possible definitions for signal and noise. We then enumerate the models used to compute the correlations between DA and SNR, and the benchmark suite.

3.1. Decision Accuracy

Decision accuracy (DA) measures whether the ranking induced by small proxy models matches the ranking of larger target models on a benchmark. Following Heineman et al. [8], we define it as

$$\mathrm{DA} = \frac{1}{|\mathscr{P}|} \sum_{(a,b)\in\mathscr{P}} \mathbb{1}\big[\mathrm{sign}(B(s_a) - B(s_b)) = \mathrm{sign}(B(m_a) - B(m_b))\big] \tag{1}$$

where $\mathscr{P}$ is a set of model pairs, s_a is the small proxy model trained with characteristic a, m_a is the corresponding target model, and $B(\cdot)$ returns the benchmark score. A higher DA means rankings observed at small scale transfer reliably to large scale, and therefore that the benchmark can guide development decisions.

Pretraining DA. For pretraining, we propose two variants. DA by size compares models from the same family across two sizes, for example 1B and 8B parameters. DA by checkpoint compares intermediate checkpoints of a single model, taken at 10%, 33%, 66%, or 100% of training in terms of FLOPS.

Post-training DA. For post-training, we also propose two variants. Cross-context DA measures whether short-context scores preserve recipe rankings after context extension. Cross-stage DA measures whether checkpoint scores preserve rankings across the SFT, DPO, and RLVR stages.

From DA to SNR. Decision accuracy is expensive to compute, since it requires evaluating the larger target models. We therefore search for cheaper benchmark statistics whose values correlate with DA, computed at the benchmark level and at the sample level, for each language and each stage. The next subsection introduces the candidates.

3.2. Signal and Noise

The signal-to-noise ratio (SNR) summarizes a benchmark's reliability as the ratio between signal, the variance of mean scores across recipes, and noise, the variance across checkpoints and seeds for a fixed recipe. We define it as

$$\mathrm{SNR} = \frac{\mathrm{Signal}}{\mathrm{Noise}}. \tag{2}$$

A higher SNR means score differences across models are large relative to the benchmark's intrinsic variability. The framework is not tied to a particular signal or noise definition, so we evaluate several candidates and select the pair with the strongest empirical correlation to DA.

Signal candidates. Given the final checkpoints of training runs at similar compute C_{final}, we consider five candidate signal definitions to capture a different notion of how spread out model scores are. Variance is the average squared distance from the mean. Mean distance is the average pairwise distance between points. Relative standard deviation is the standard deviation divided by the mean. Star discrepancy is the largest difference between any point and the uniform distribution. Dispersion is the largest pairwise distance,

$$\mathrm{Dispersion}(C_{\mathrm{final}}) = \max_{i \neq j} \|c_i - c_j\|. \tag{3}$$

Noise candidates. We consider four sources of noise that the framework can in principle account for. Seed noise is the the standard deviation of the final checkpoint across training runs with different random seeds. Data order noise is the equivalent across runs that vary the order of training documents. Total variation is the average change in metric score across training checkpoints minus an improvement term. Checkpoint-to-checkpoint noise is proposed by Heineman et al. [8] as a proxy when the first three are too expensive to estimate at large scale and depends only on the final n training checkpoints:

$$\text{Checkpoint noise} = \sigma\big(\{U(t_j)\}_{j=T-k+1}^{T}\big). \tag{4}$$

We propose a fifth noise definition, computable from a single evaluation run on any public model. Benchmark noise is the average relative standard deviation of scores across $k = 5$ disjoint folds of the evaluation set:

$$\mathrm{Noise}(M) = \frac{1}{|M|} \sum_{m \in M} \frac{\sqrt{\frac{1}{k}\sum_{i=1}^{k}(m_i - \bar{m})^2}}{\bar{m}} \tag{5}$$

where $m_1, \ldots, m_k$ are the per-fold scores of model m. The fold scores are obtained by partitioning the stored per-question outputs of a single evaluation run, so no inference is repeated and no intermediate checkpoints are needed. This makes the framework applicable to any model with public scores.

3.3. Models

We pretrain 36 small models across four scales, 175M, 350M, 600M, and 1B non-embedding parameters, using three English/multilingual data mixtures (30/70, 60/40, and 90/10) and three random seeds each. We extend the analysis to 8B and 70B models by considering open-source models with available intermediate checkpoints.

Custom proxy models. All models follow a decoder-only transformer architecture with grouped query attention, rotary positional embeddings (RoPE) and XiELU as the activation function. The architecture scales by jointly increasing depth, model dimension, and number of attention heads while keeping the KV head ratio fixed at 1/4. All models are trained on approximately 100B tokens drawn from FineEdu-DCLM and FineWeb2 [34], with a shared vocabulary of 131,072 tokens. Since this vocabulary size places a large fraction of parameters in the embeddings at these scales, we tie input and output embeddings. Training is done in BF16 mixed precision using Megatron-LM [35].

Data mixtures. We construct 3 mixtures by varying the ratio of English to multilingual data, namely 90/10, 60/40, and 30/70. The English component comes from FineWeb-Edu [36] filtered through the DCLM pipeline [37]. The multilingual component is drawn from FineWeb2 [34] using the Apertus high-quality filter [38], covering the top 200 languages and preserving the original language distribution of the corpus.

Open-source models. We extend the analysis to larger scales and later training stages with three open-source multilingual model families (Apertus [39], SmolLM3 [40], and OLMo3 [41]) with intermediate checkpoints across post-training stages and sizes from 0.6B to 8B.

3.4. Benchmarks

Languages. We evaluate on 12 languages: English, Spanish, Russian, Hindi, Chinese, Japanese, Arabic, Turkish, Thai, Vietnamese, Swahili, and Basque. The selection balances typological breadth with benchmark availability. We include languages from across the resource spectrum, covering seven language families and one language isolate [42], six writing systems, and typological features that stress tokenization and modeling [22, 23, 43].

Table 1
Language selection. We cover seven language families plus one isolate, six writing systems, and high to low-resource.

Code	Language	Family	Script	Resource
en	English	Indo-European, Germanic	Latin	High
es	Spanish	Indo-European, Romance	Latin	High
ru	Russian	Indo-European, Slavic	Cyrillic	High
hi	Hindi	Indo-European, Indo-Aryan	Devanagari	Mid
zh	Mandarin Chinese	Sino-Tibetan	Han	High
ja	Japanese	Japonic	Kanji and Kana	High
ar	Modern Std. Arabic	Afro-Asiatic, Semitic	Arabic	High
vi	Vietnamese	Austroasiatic, Vietic	Latin	Mid
tr	Turkish	Turkic	Latin	Mid
th	Thai	Tai-Kadai	Thai	Mid
sw	Swahili	Niger-Congo, Bantu	Latin	Low
eu	Basque	Isolate	Latin	Low

Benchmark suite. We evaluate pretraining benchmarks covering several capabilities: world knowledge, reading comprehension, natural language inference, commonsense reasoning (decomposed into physical, narrative, and activity-based variants), grammatical acceptability, and truthfulness [44, 45, 46]. The mid-training suite reuses some pretraining benchmarks and focuses on regional knowledge, reasoning capabilities, including math, code, and long-context tasks [20, 21, 47]. The post-training suite considers reasoning, code, and adds instruction-following, bias, and safety benchmarks [48, 49, 50].

4. Proposed Experiments and Analysis

We organize the experimental program around three goals, namely measuring benchmark reliability on the custom models (4.1), testing whether the resulting benchmark selection generalizes (4.2), and analyzing which benchmark design choices drive higher reliability (4.3).

4.1. Computing SNR and Decision Accuracy

Checkpoint selection. For each custom and open-source model, we evaluate 10 evenly spaced checkpoints across training and 5 additional checkpoints in the final 10% of training FLOPS. This

Table 2
Benchmark suite. Stage indicates the training stage at which each benchmark is used: pre (pretraining), mid (mid-training), post (post-training).

Benchmark	Capability	Languages	Stage
`arc`	scientific knowledge	ar, en, es, eu, hi, ru, vi, zh	pre, mid, post
`belebele`	reading comprehension	ar, en, es, eu, hi, ja, ru, sw, vi, zh	pre
`global_mmlu`	world knowledge	ar, en, es, hi, ja, ru, sw, tr, vi, zh	pre, mid, post
`global_piqa`	physical commonsense	ar, en, es, hi, ja, ru, sw, tr, vi, zh	pre
`hellaswag`	activity commonsense	ar, en, es, hi, ru, tr, vi, zh	pre, mid, post
`mlqa`	cross-lingual QA	ar, en, es, hi, vi, zh	pre
`mgsm_direct`	mathematical reasoning	en, es, eu, ja, ru, sw, zh	pre
`multiblimp`	grammatical acceptability	ar, en, es, eu, hi, ru, tr	pre
`truthfulqa_mc1`	truthfulness	en, es, ja, ru, sw, zh	pre, mid
`xnli`	natural language inference	ar, en, es, eu, hi, ru, sw, tr, vi, zh	pre
`xstorycloze`	narrative commonsense	ar, en, es, eu, hi, ru, sw, zh	pre
`include_base_44`	regional knowledge	ar, en, es, hi, ja, ru, sw, tr, vi, zh	mid, post
`mgsm_cot`	mathematical reasoning	en, es, ja, ru, sw, zh	mid, post
`mlogiqa`	logical reasoning	ar, en, es, hi, ru, vi, zh	post
`mifeval`	instruction following	ar, en, es, hi, ja, ru, tr, vi, zh	post
`multi_if`	instruction following	en, es, hi, ja, ru, tr, vi, zh	post
`pmmeval`	multitask reasoning	ar, en, es, ja, ru, tr, vi, zh	post
`crows_pairs`	social bias	en, es, ar, hi, ru, zh	post

schedule captures both the long-range trajectory and the late-training variation.

SNR computation. We compute the signal-to-noise ratio under each of the signal and noise definitions introduced in Section 3, for every benchmark and every language. This lets us identify which definition correlates most strongly with decision accuracy for each language.

Correlation with decision accuracy. We split the 36 custom models into a 24 model fitting set and a 12 model held-out set. On the fitting set, we compute decision accuracy by size, using the ranking from a small model to predict the ranking of a larger one in the same family, and decision accuracy by checkpoint, using a single early checkpoint at 10%, 33%, and 50% of training FLOPS to predict the final ranking. We then test on the held-out set whether the benchmarks identified as high SNR retain high decision accuracy. This is the central validation of the framework.

Language-dependent metric. The same benchmark may be reliable for some languages and noisy for others. We therefore report all SNR and decision accuracy values per language and aggregate into a benchmark reliability map that practitioners can consult for the languages they target.

4.2. Framework Generalization

Extrapolation to new languages. We test whether the benchmarks identified as reliable for our 10 languages remain reliable on languages outside the test set. We first consider languages typologically close to the test set, namely French, German, and Ukrainian, and then more distant languages such as Vietnamese and Swahili. For each new target language, we compute SNR and decision accuracy, and report which cases preserve the expected high decision accuracy and which fail. This identifies the limits of the language-based reliability map.

Extrapolation to other design decisions. The original experiments vary the multilingual data mixture. We rerun the same protocol with custom models from 175M to 1B parameters that share data mixtures but use four different tokenizers, namely a parity-aware tokenizer, a SuperBPE tokenizer, and two standard BPE tokenizers with English-heavy and multilingual vocabularies. This tests whether the framework supports design decisions beyond data mixture.

Comparison with other benchmark validity frameworks. We compare against three reference frameworks: the English SNR of Heineman et al. [8], FineTasks [51], and SMART filtering [16]. For each, we identify the benchmarks reported as most informative, compute their SNR and decision accuracy on our model suite using checkpoints under the same schedule, and analyze the overlap between validity definitions. We also report the compute cost of each framework, which is essential for practical adoption.

4.3. Improving Benchmark Quality

Benchmark creation decisions. We document the data source and curation method of each benchmark in our suite, and compute the correlation between these characteristics and SNR. We classify the data source as web data, news, literature, crowdsourced, expert-generated, or synthetic, and the curation method by whether the data was machine or manually translated and by the review process applied. This yields a recipe for high-SNR benchmark construction.

Subsampling methods. We then explore subsampling rules that increase SNR on existing benchmarks, including selecting harder questions and removing semantically similar items. The output is a set of higher-SNR subsets that practitioners can use as replacements for full benchmark runs when running ablations.

5. Discussion

We list open discussion points that would help us prioritize the experiments and guide the analysis.

Signal and noise definition selection. We propose candidate signal and noise measures, and plan to select the pair with the highest correlation to decision accuracy. Fitting both choices on the same data risks overfitting the reliability map. We would like to discuss whether a held-out language set, a held-out benchmark set, or a combination is the right protection, and how to report the selected pair honestly.

Language coverage and the generalization claim. The framework will be calibrated on 12 languages and validated on a small set of held-out languages including French, German, Ukrainian, Vietnamese, and Swahili. The strength of the resulting generalization claim depends on how the calibration set is described. We would like input on whether a typological feature based stratification would support a stronger claim than the resource-level stratification we currently use.

Subsampling methodology. To improve benchmark reliability we plan to explore item selection rules such as removing easy or semantically redundant questions. SMART [16] and item response theory based approaches [11] offer two routes. We would like to discuss whether to commit to one method, run a head to head comparison, or develop a multilingual-specific variant that conditions on language and script.

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